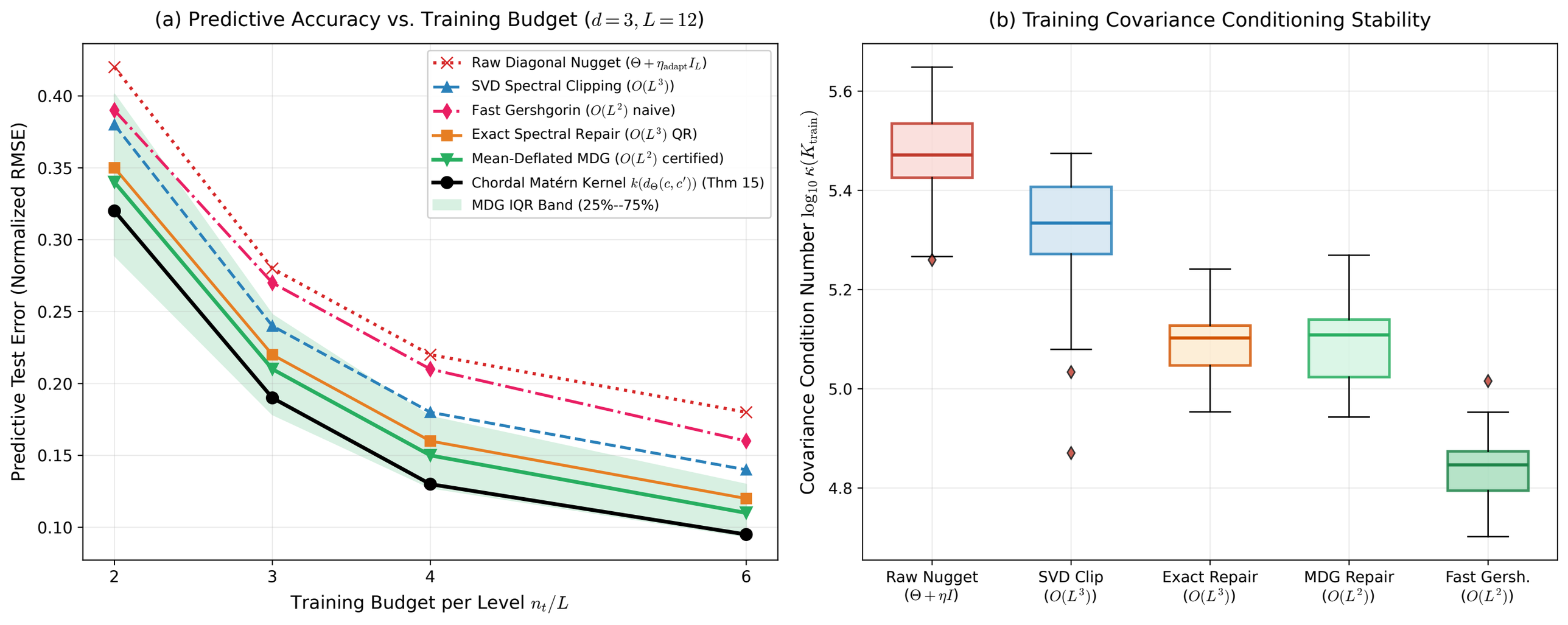
(a) Predictive Accuracy vs. Training Budget ($d = 3, L = 12$)
Predictive Test Error (Normalized RMSE)
Training Budget per Level n_t/L
Raw Diagonal Nugget ($\Theta + \eta_{\text{adapt}} I_L$)
SVD Spectral Clipping ($O(L^3)$)
Fast Gershgorin ($O(L^2)$ naive)
Exact Spectral Repair ($O(L^3)$ QR)
Mean-Deflated MDG ($O(L^2)$ certified)
Chordal Matérn Kernel $k(d_\Theta(c, c'))$ (Thm 15)
MDG IQR Band (25%--75%)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
2
3
4
6
(b) Training Covariance Conditioning Stability
Covariance Condition Number $\log_{10} \kappa(K_{\text{train}})$
5.6
5.4
5.2
5.0
4.8
Raw Nugget ($\Theta + \eta I$)
SVD Clip ($O(L^3)$)
Exact Repair ($O(L^3)$)
MDG Repair ($O(L^2)$)
Fast Gersh. ($O(L^2)$)